

Thesis paper Synopsis

On

**Comparative Study on Plastic Pollution in Two Tourist Destination from
Rangamati and Cox's Bazar**

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Background of the study

Plastic pollution is a growing environmental concern, particularly in regions heavily dependent on tourism. Tourist destinations, while contributing significantly to local economies, often face severe environmental degradation due to improper waste management. In coastal and hill regions, plastic waste not only disrupts natural ecosystems but also threatens the livelihoods of communities reliant on these ecosystems for fishing, agriculture, and tourism. Globally, over 8 million tons of plastic waste enter the oceans annually, impacting marine life and coastal habitats (Jambeck et al., 2015). Similarly, terrestrial ecosystems in hill tracts suffer from the accumulation of non-biodegradable waste, impacting biodiversity and soil health (UNEP, 2018).

Rangamati, situated in the Chittagong Hill Tracts, is renowned for its pristine lakes and rich cultural heritage, attracting thousands of domestic and international tourists annually. However, increased tourist activity has led to a rise in plastic waste, particularly in and around popular spots like Kaptai Lake, threatening aquatic biodiversity and disrupting local ecosystems (Roy et al., 2021). Inadequate waste disposal systems exacerbate the problem, as most plastic waste ends up in water bodies or is burned, releasing harmful pollutants into the air.

On the other hand, Cox's Bazar, home to the world's longest sandy beach, is one of Bangladesh's most visited destinations, drawing millions of tourists each year. While tourism boosts the local economy, the high influx of visitors results in substantial plastic waste generation, including single-use plastics and food packaging. This pollution poses severe threats to marine ecosystems and species like sea turtles, which are already endangered (Islam & Tanaka, 2020). Furthermore, plastic waste in coastal regions often enters the ocean through tides, contributing to global marine pollution.

Comparing these two regions provides insights into the varying impacts of tourism-driven plastic pollution in hill and coastal ecosystems. Identifying the differences in waste management practices and the extent of pollution in Rangamati and Cox's Bazar can inform sustainable tourism policies and targeted waste management strategies to mitigate environmental degradation.

Aims and objectives

The goal of the study is to evaluate the extent of plastics pollution in two popular destinations of Bangladesh, Rangamati and Cox's Bazar. However, the specific objectives of the study are:

1. Identify the abundance of different plastics in the two tourist spots.
2. To characterize the plastics from collected samples.

Methodology

Study area

The study will be conducted on various sampling stations along the coastal waters of Cox Bazar, Bangladesh and Rangamati, Bangladesh during the time between January 2025 to April 2025. Ten sampling locations will be chosen as the representative of entire Cox's Bazar cost and Rangamati.

Sampling method

At the sampling site (in Cox's Bazar) I will choose an area with enough space to apply the intended sampling plan for monitoring plastic marine debris. Sampling will be done on the three parallel lines including high strandline, intermediate and backshore on each sampling station. 9 quadrats (1x1 m²) will be sampled from each sampling station. In Rangamati the sample will be collected directly from the surface.

Sample collection

After taking all the necessary equipment, all the plastic objects will be seen on the sand surface of each quadrant and will be picked up using tweezers (in Cox's Bazar). Then sand will be collected from the top 10 cm from each quadrat. In Rangamati the sample will be collected from the surface area.

Sample processing

Sample will be taken into the lab. Dry samples will be sieved immediately but wet samples will be air dried before sieving.

Data analysis

Quantification of Plastics: Categorize and quantify collected plastics by size, type, and source (e.g., fishing gear, packaging).

Microplastic Analysis: Use laboratory techniques like microscopy and spectroscopy to identify and quantify microplastics in water, sediment, and biological samples.

Statistical analysis: Compare the abundance of recreational and non-recreational beaches Welch two sample t test Mann-Whitney U test will be employed All analysis will be done in the R studio.

Ecological Impact Assessment: Evaluate the impacts of plastic pollution on fish diversity, abundance, and health indicators.

Expected results

This research will identify the sources of plastic pollution in Cox's Bazar Beach and Kaptai Lake, assess its impact on aquatic biodiversity and fish populations, and examine the risks of microplastics entering the food chain. It will also explore the socio-economic effects on fisheries-dependent communities. The study will propose strategies for reducing plastic waste and promoting sustainable management practices, providing baseline data for future research and long-term solutions to protect these ecosystems.

References

- Islam, M. S., & Tanaka, M. (2020). Impacts of pollution on coastal and marine ecosystems in the Bay of Bengal region. *Marine Pollution Bulletin*, 160, 111655.
- Jambeck, J. R., et al. (2015). Plastic waste inputs from land into the ocean. *Science*, 347(6223), 768-771.
- Roy, R., et al. (2021). Environmental impacts of tourism in the Chittagong Hill Tracts: A review. *Journal of Environmental Research*, 8(3), 56-63.
- UNEP (2018). Single-use plastics: A roadmap for sustainability. United Nations Environment Programme.